Mansoura University
Faculty of Computers and Information
Department of Information System
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Agile Methods

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Lecture 9
Data and Process Modeling (DFD)

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DATA FLOW Diagram

Logical Vs Physical Models

Data and Process Modeling Tools

Data Flow Diagrams and Symbols

Creating a set of DFDs

- Guidelines for drawing DFDs
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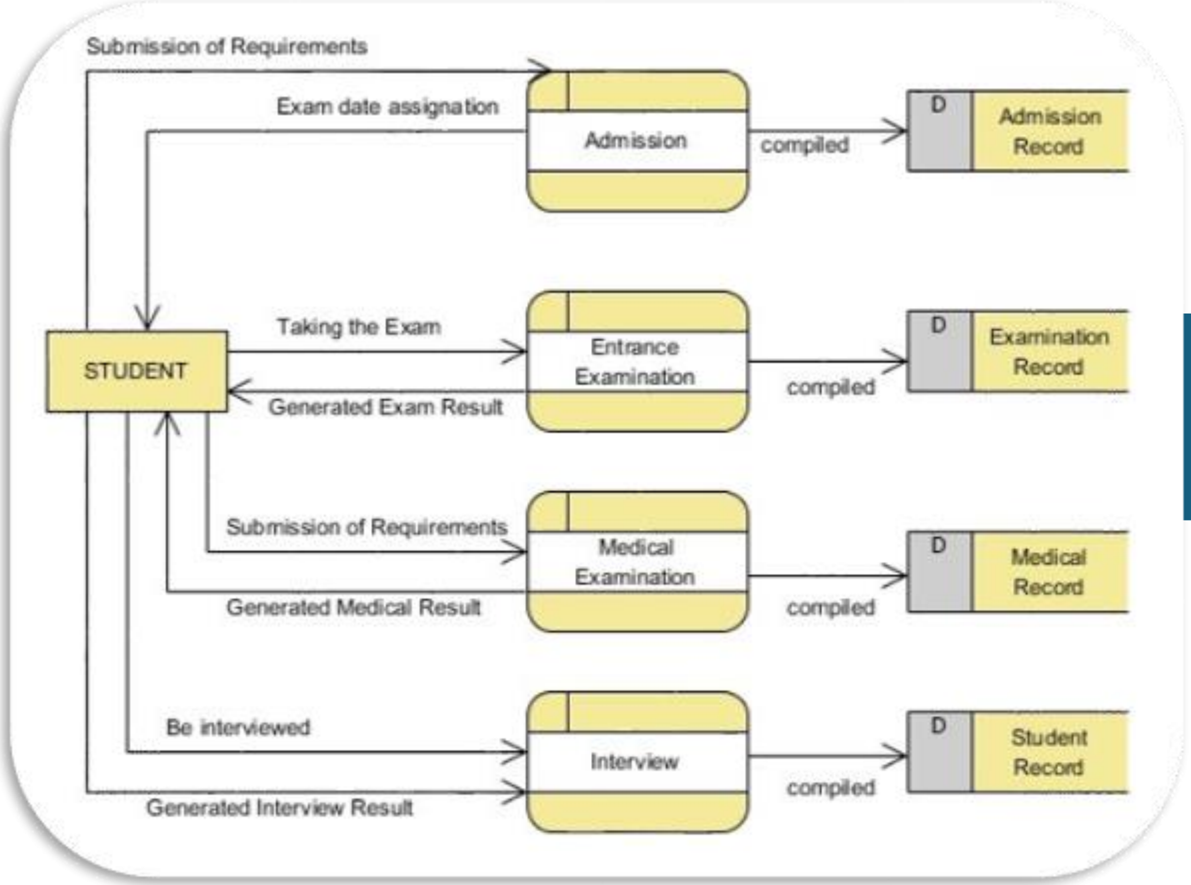
  - Draw a DFD diagram 0
  - Functional Primitives
  - 0 Diagram exploding can be enough?



Logical Vs Physical Model

The Requirement modeling from structure or O-O or agile perspectives is just a logical model

logical model

- shows *what* the system must do., regardless of how it will be implemented physically.
- Analysis Phase Based
- DFD, (Personas and User Stories), Class Diagram

Physical Model

- **A physical model** is built that describes *how* the system will be constructed.
- Design Phase Based
- ERD, System Architecture, Network Architecture

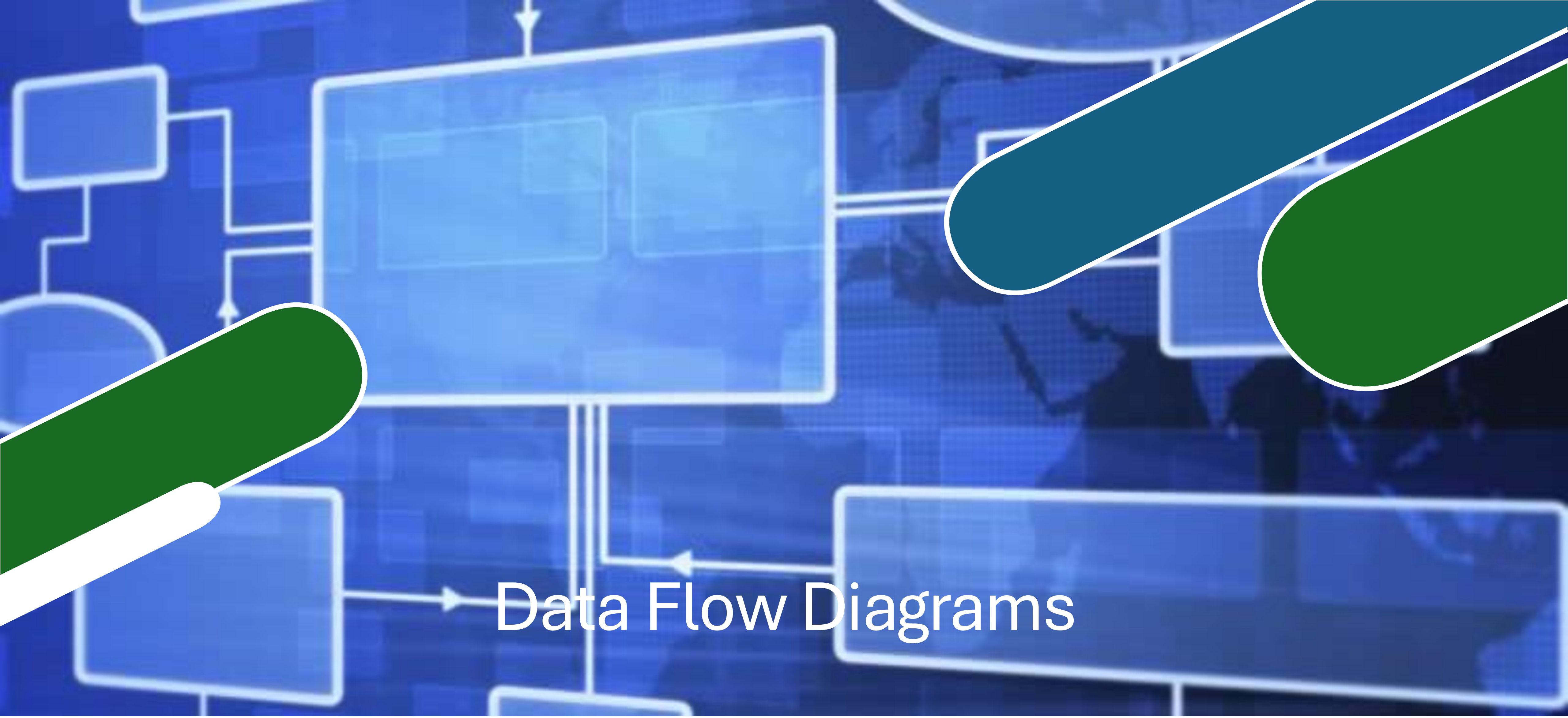
DATA AND PROCESS MODELING TOOLS

Systems analysts use many graphical techniques to describe an information system.

One popular method is to draw a set of data flow diagrams.

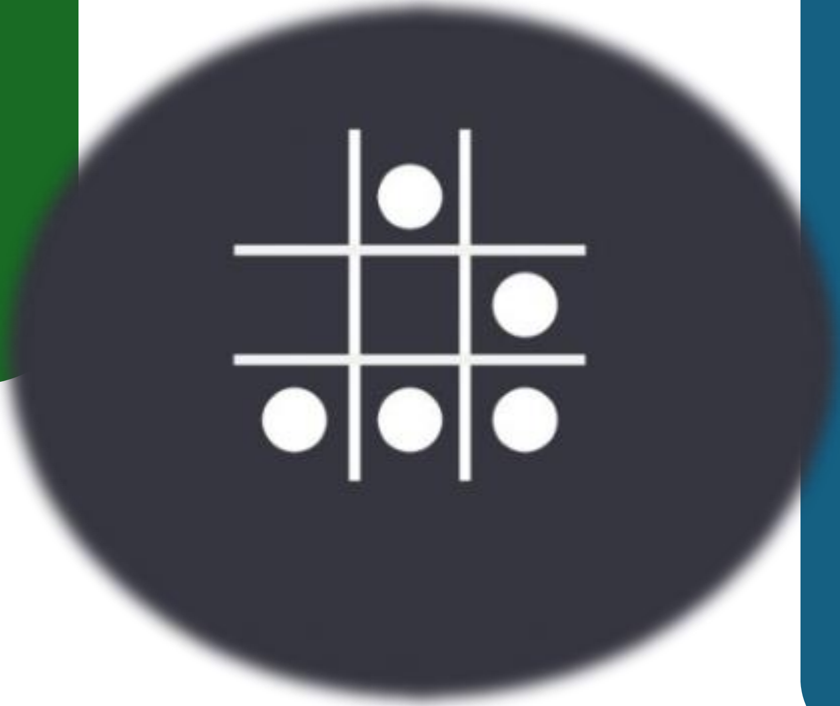
A data flow diagram (DFD) uses various symbols to show how the system transforms input data into useful information.

Other graphical tools include object model.



Data Flow Diagrams

A DFD shows how data moves through an information system but does not show program logic or processing steps.



DATA Flow Diagram

A set of DFDs provides a logical model that shows what the system does, not how it does it.

That distinction is important because focusing on implementation issues at this point would restrict the search for the most effective system design.

DFD SYMPOLS

DFDs use four basic symbols that represent processes, data flows, data stores, and entities.

Several different versions of DFD symbols exist, but they all serve the same purpose.

- **Gane and Sarson**
- **Yourdon and Coad**

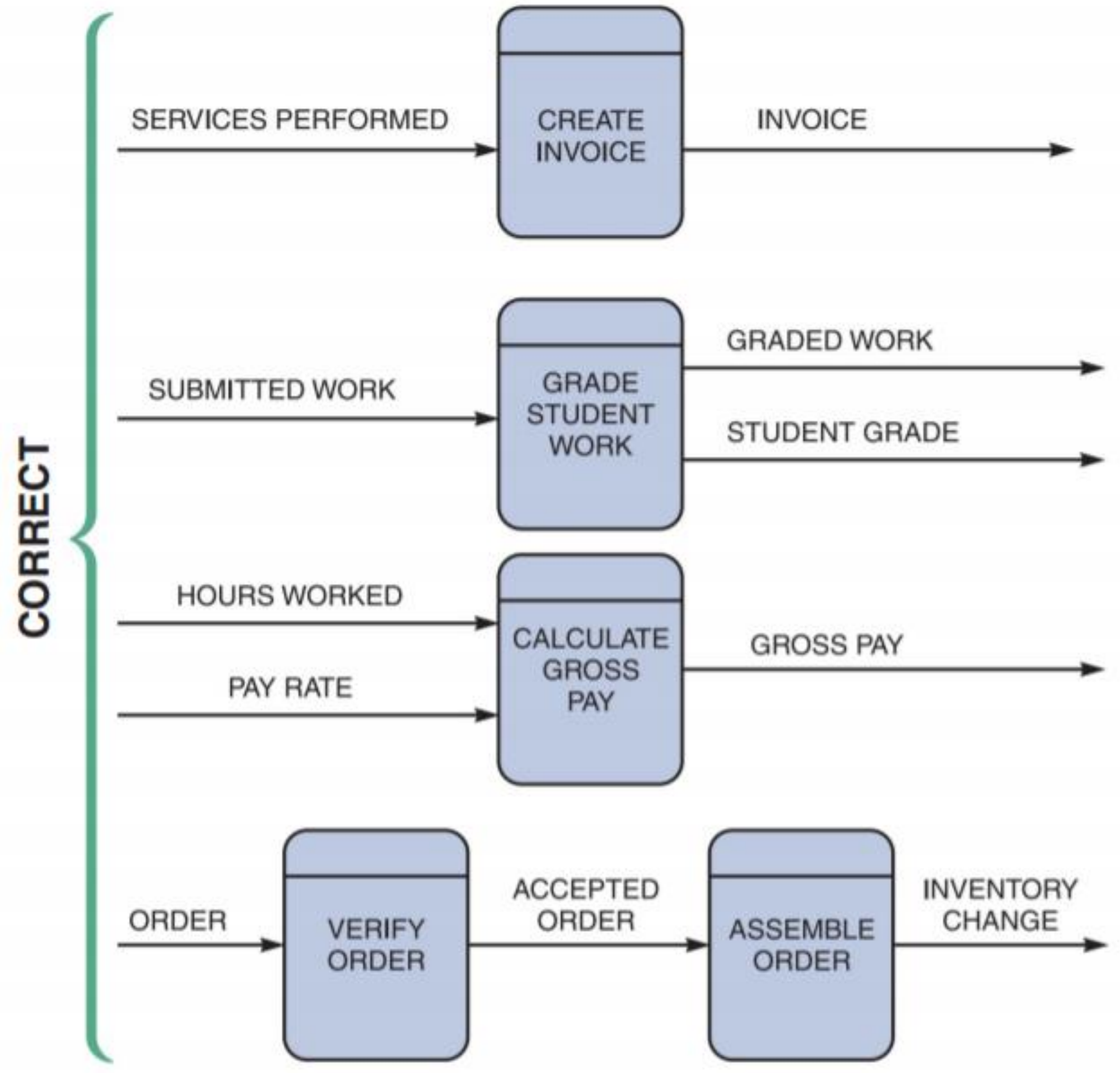
Data Flow Symbol

- A data flow is a **path** for data to move from one part of the information system to another.
- A data flow in a DFD represents one or more data items.

Correct Data Flaws

- Any service in world or process have a component of a data entry, and a component for resulting the output.

- The process must have data flows for input and output for each same process. and must be a logical flows, not a magic process.



Incorrect Data Flaws

- SPONTANEOUS GENERATION.

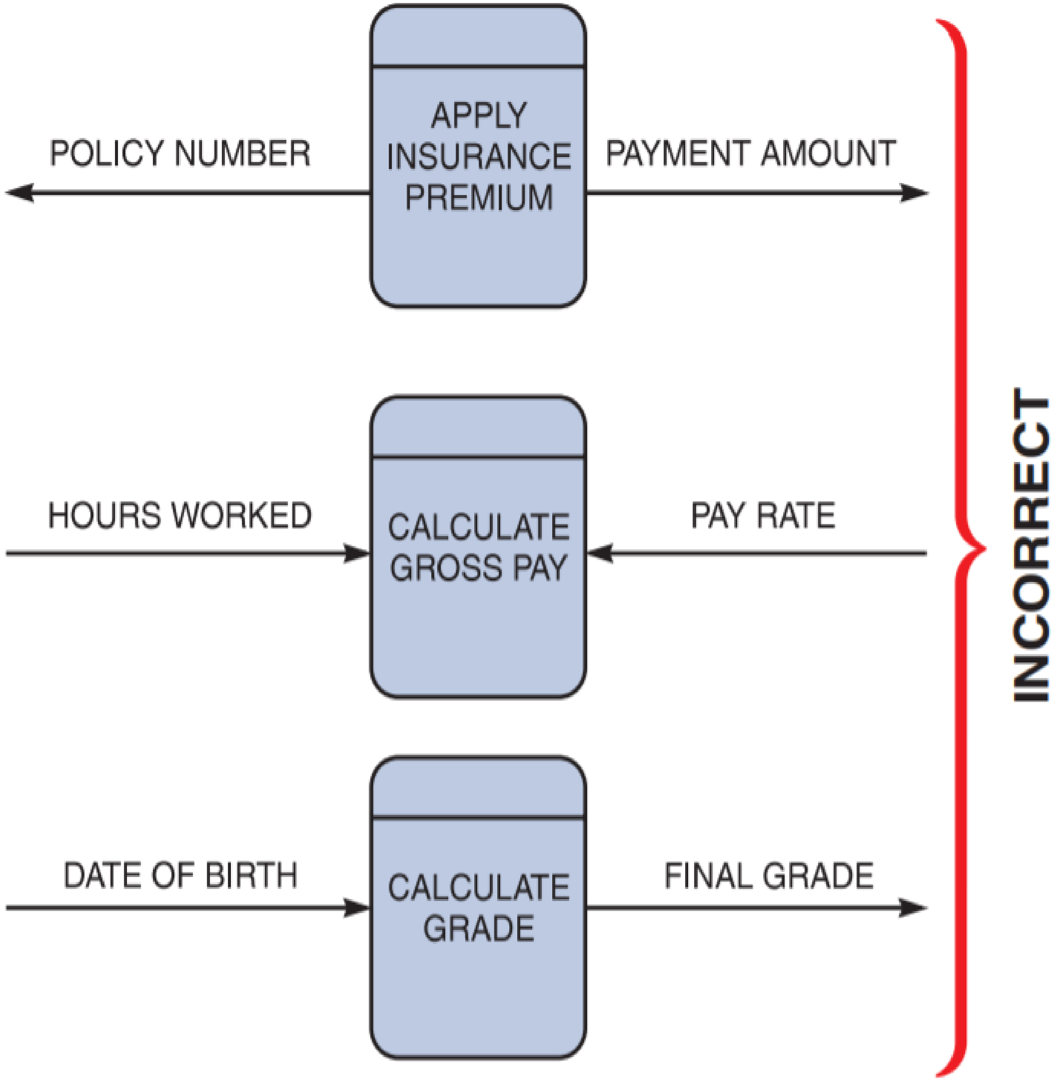
  ➢ The APPLY INSURANCE PREMIUM process, for instance, produces output, but has no input data flow. Because it has no input, the process is called a spontaneous generation process.

- **BLACK HOLE.**

  ➢ CALCULATE GROSS PAY is called a black hole process, which is a process that has **input**, but produces no output.

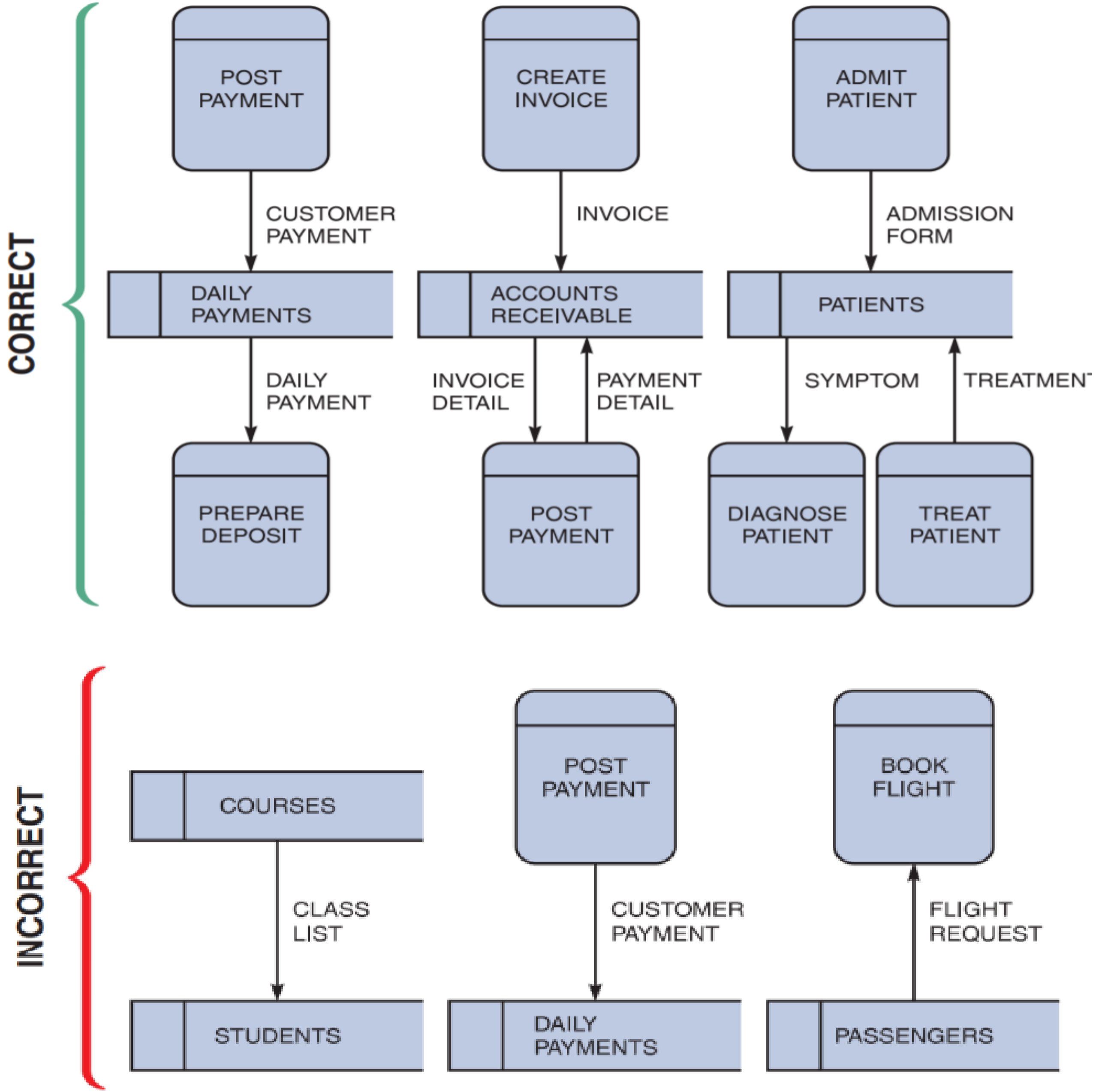
- **GRAY HOLE.**

  ➢ A gray hole is a process that has at least one input and one output, but the input obviously is **insufficient to generate the output shown**.



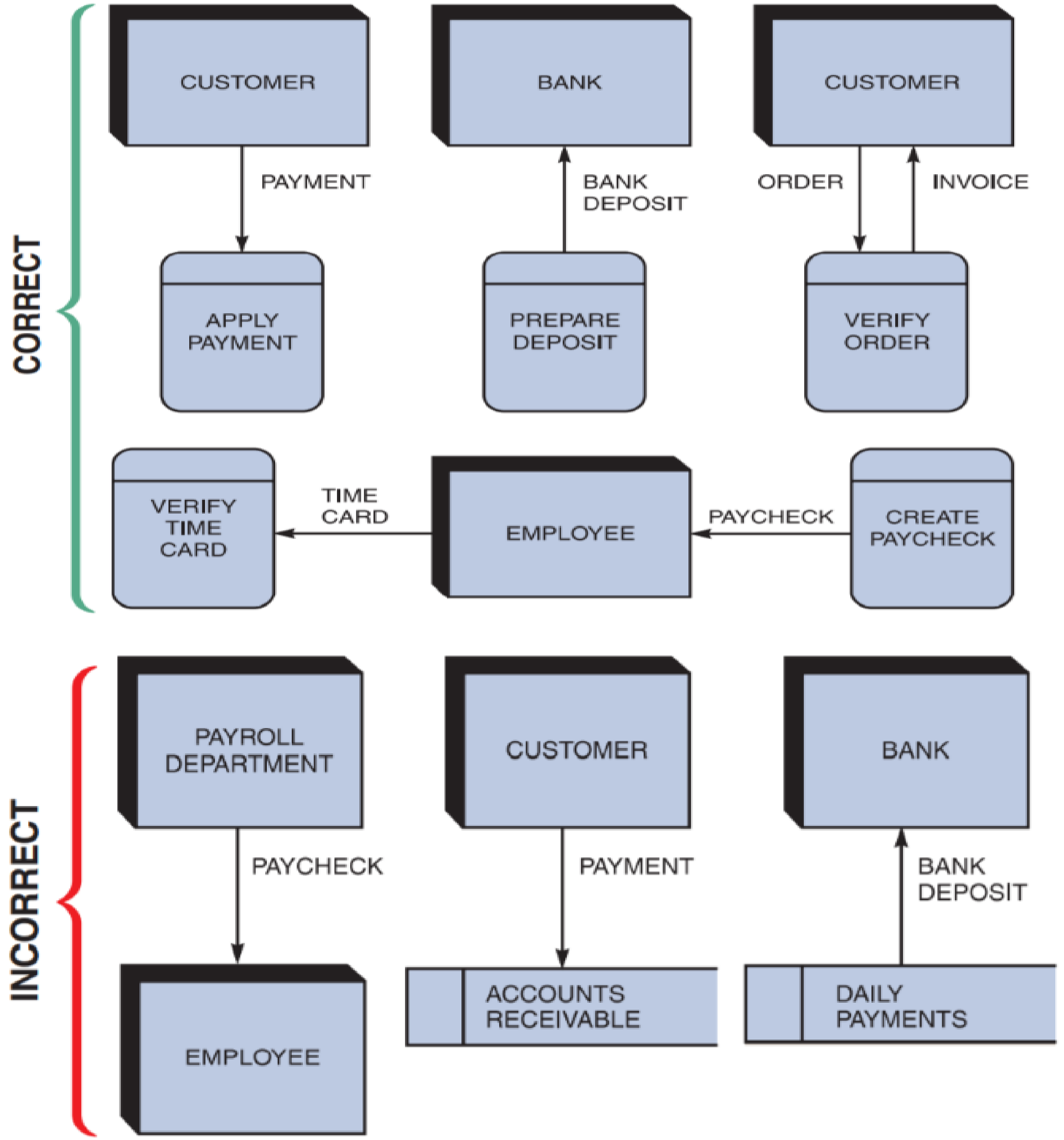
DATA STORE SYMBOL

- A data store is used in a DFD to represent data that the system stores because one or more processes need to use the data at a later time.
- A data store must be connected to a process with a data flow.
- There is an exception to the requirement that a data store must have at least one incoming and one outgoing data flow.



ENTITY SYMBOL:

- A DFD shows only external entities that provide data to the system or receive output from the system.
- A DFD shows the boundaries of the system and how the system interfaces with the outside world.
- DFD entities are also called terminators because they are data origins or final destinations.



Correct and Incorrect Examples of Data Flows

	Process to Process	✓
	Process to External Entity	✓
	Process to Data Store	✓
	External Entity to External Entity	✗
	External Entity to Data Store	✗
	Data Store to Data Store	✗

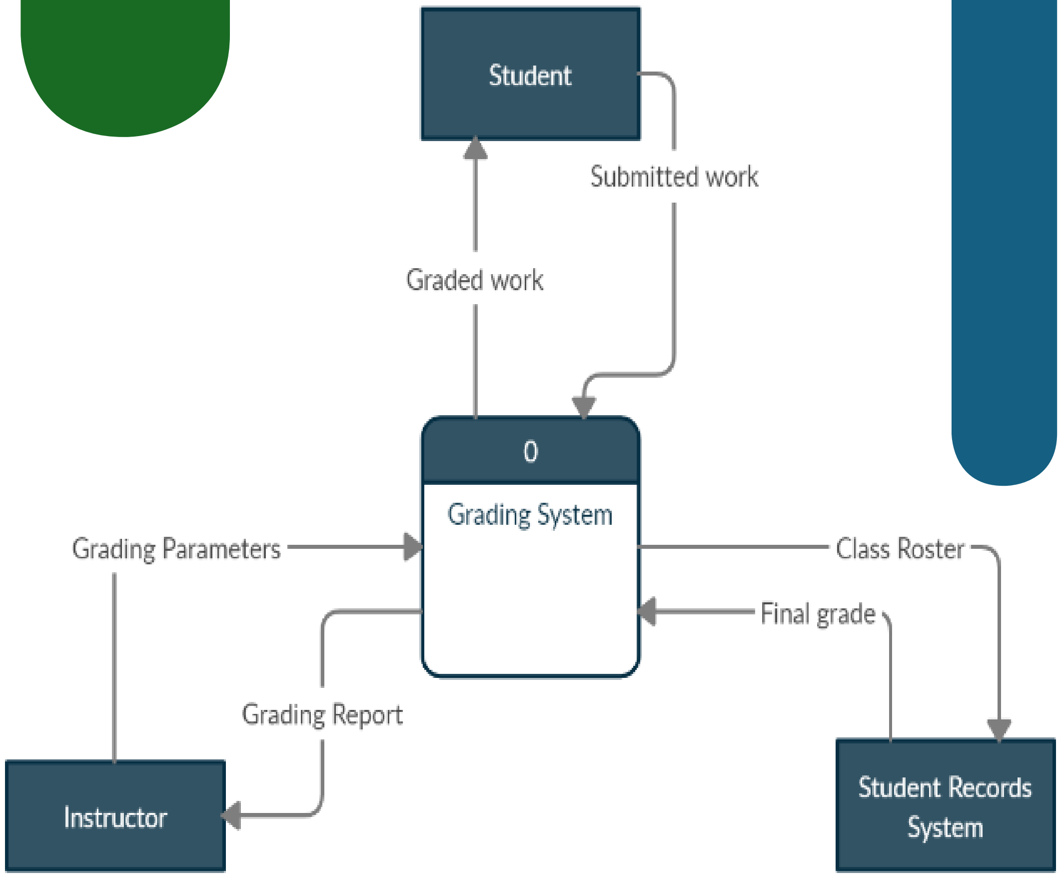
CREATING A SET OF DFDS

During requirements modeling, interviews, questionnaires, and other techniques were used to gather facts about the system, and it was explained how the various people, departments, data, and processes fit together to support business operations.

Now a graphical model of the information system is created based on the fact-finding results.

GUIDELINES FOR DRAWING DFDS

When drawing a context diagram and other DFDs, these guidelines should be followed:



DFD CONTEXT DIAGRAM

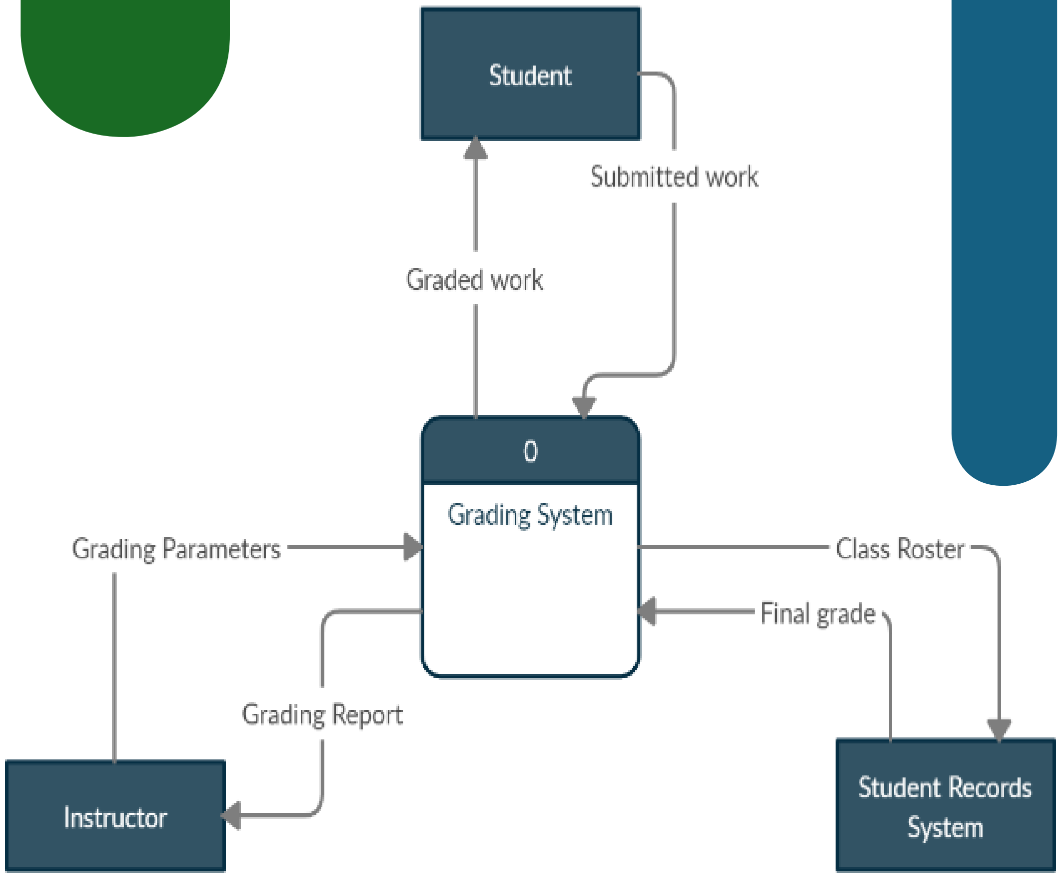
1. Draw the context diagram so it fits on one page.

2. Use the name of the information system as the process name in the context diagram.

For example, the process name is GRADING SYSTEM.

Notice that the process name is the same as the system name.

DFD CONTEXT DIAGRAM

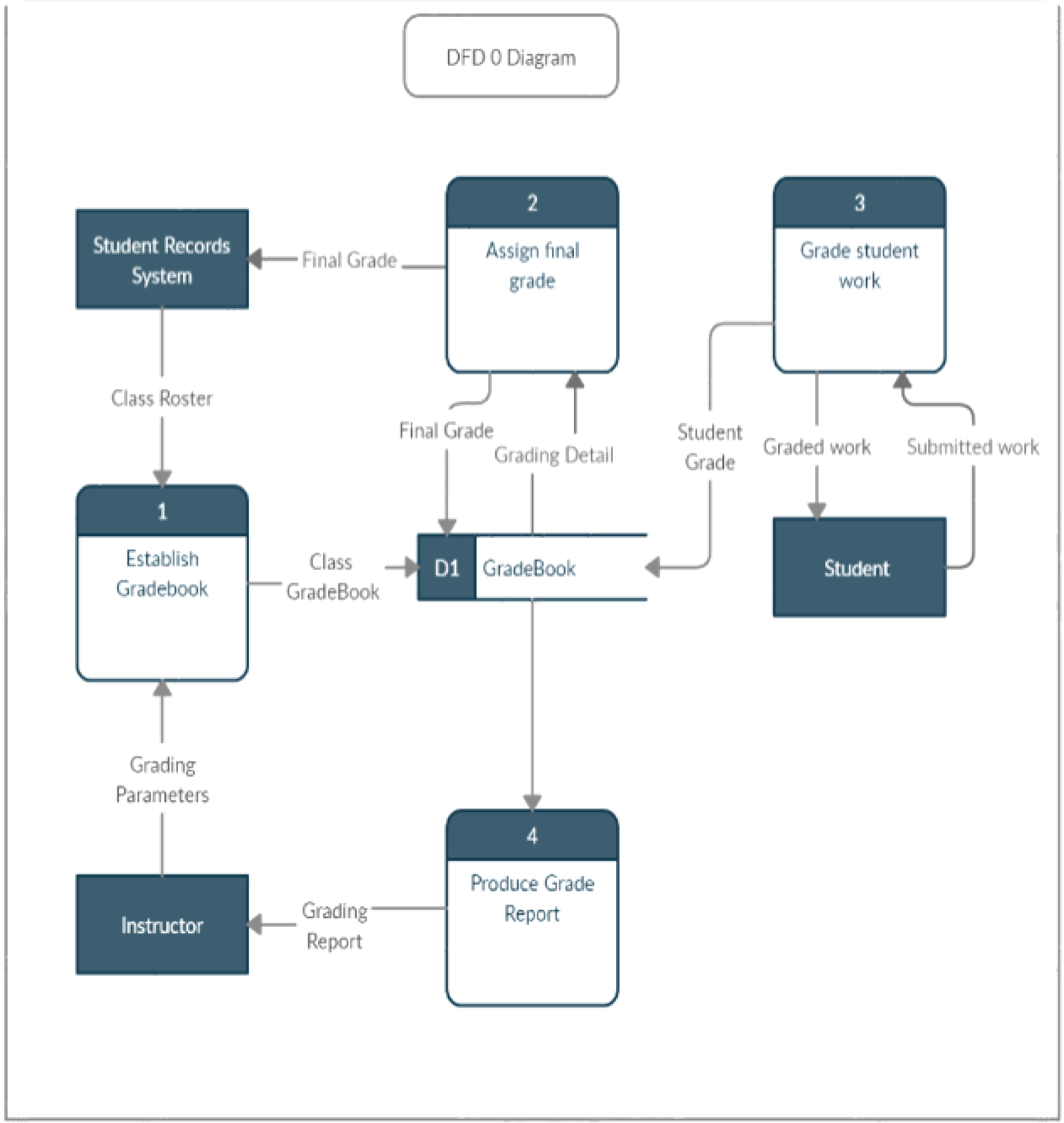


3- Use unique names within each set of symbols.

4- Do not cross lines. One way to achieve that goal is to restrict the number of symbols in any DFD.

5- Provide a unique name and reference number for each process.

6- Obtain as much user input and feedback as possible.

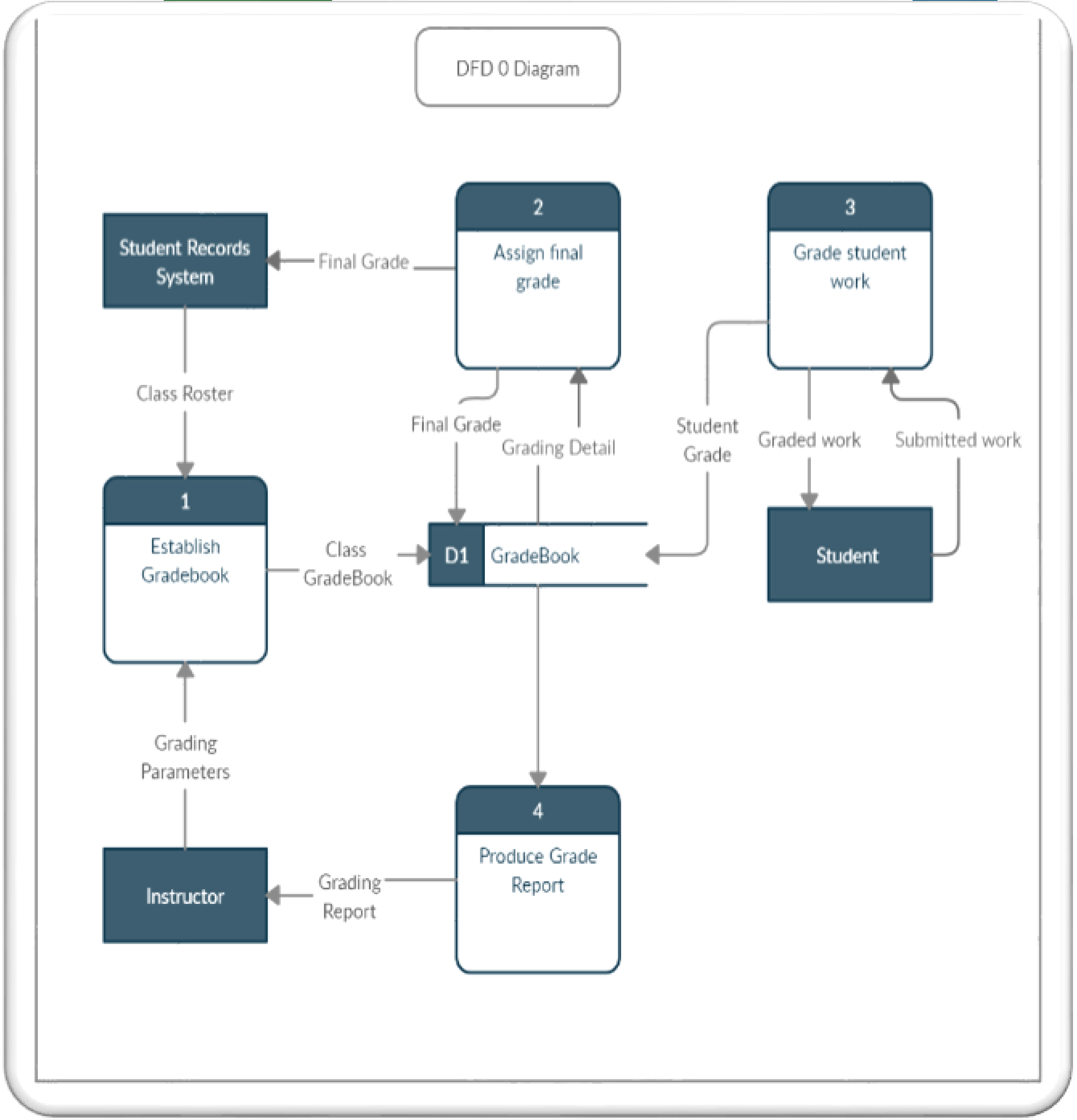


DRAW A DFD DIAGRAM 0

To show the detail inside context diagram, a DFD diagram 0 is created.

Diagram 0 (the numeral zero, and not the letter O) provides an overview of all the components that interact to form the overall system.

It zooms in on the system and shows major internal processes, data flows, and data stores.

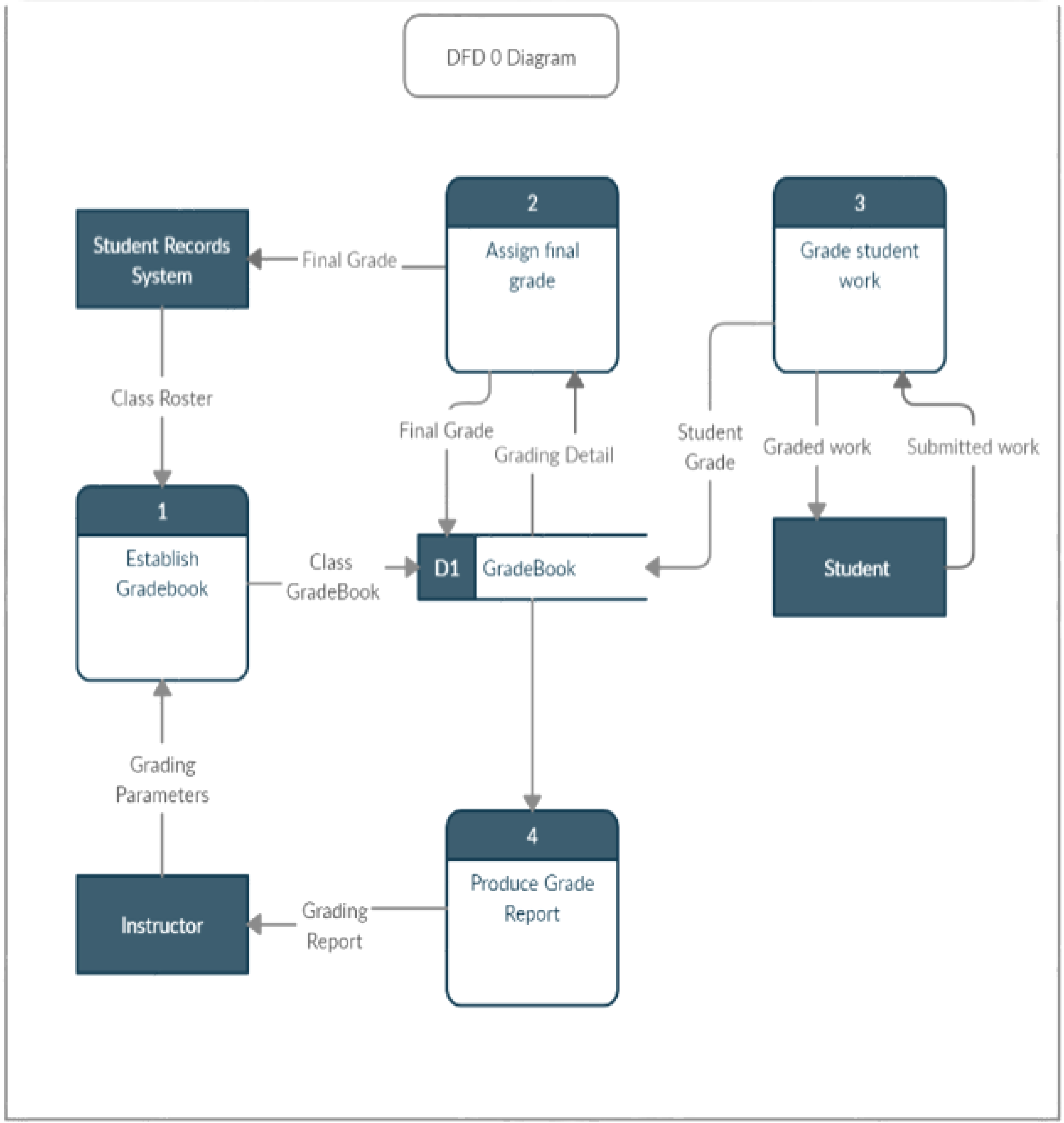


DRAW A DFD DIAGRAM 0

Notice that each process in diagram 0 has a reference number: ESTABLISH GRADEBOOK is 1, ASSIGN FINAL GRADE is 2, GRADE STUDENT WORK is 3, and PRODUCE GRADE REPORT is 4. These reference numbers are important.

Because diagram 0 is an exploded version of process 0, it shows considerably more detail than the context diagram.

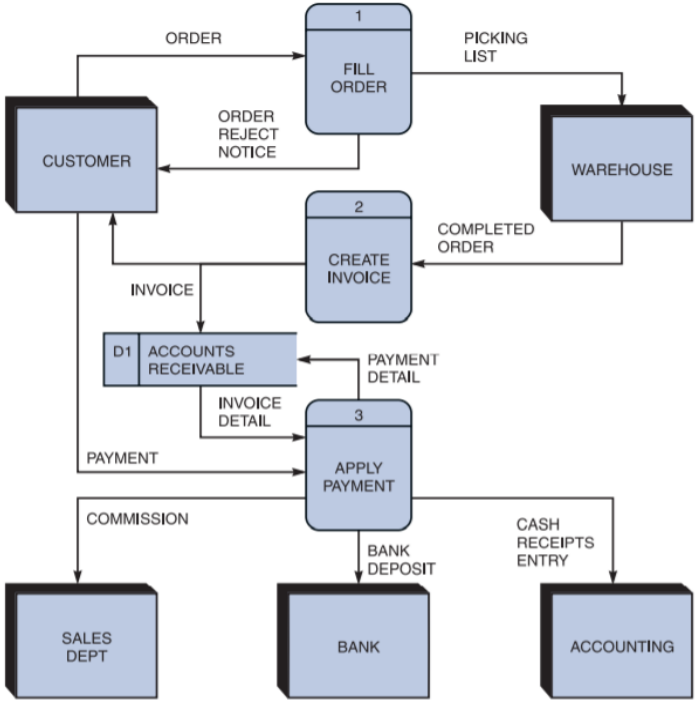
Diagram 0 can also be referred to as a partitioned or decomposed view of process 0.



DRAW A DFD DIAGRAM 0

When a DFD is exploded, the higher-level diagram is called the parent diagram, and the lower-level diagram is referred to as the child diagram.

The grading system is simple enough that no additional DFDs are needed to model the system. At that point, the four processes, the one data store, and the 10 data flows can be documented in the data dictionary.



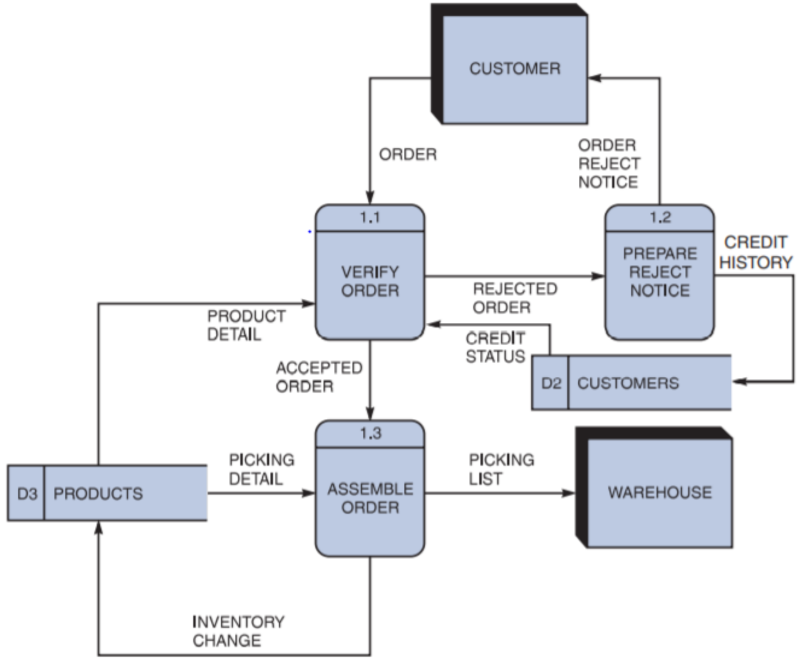


DIAGRAM EXPLODING CAN'T BE ENOUGH

This set of lower-level DFDs is based on the order system. To create lower-level diagrams, leveling and balancing techniques must be used.

Leveling is the process of drawing a series of increasingly detailed diagrams, until all functional primitives are identified.

Balancing maintains consistency among a set of DFDs by ensuring that input and output data flows align properly.

THANK YOU!